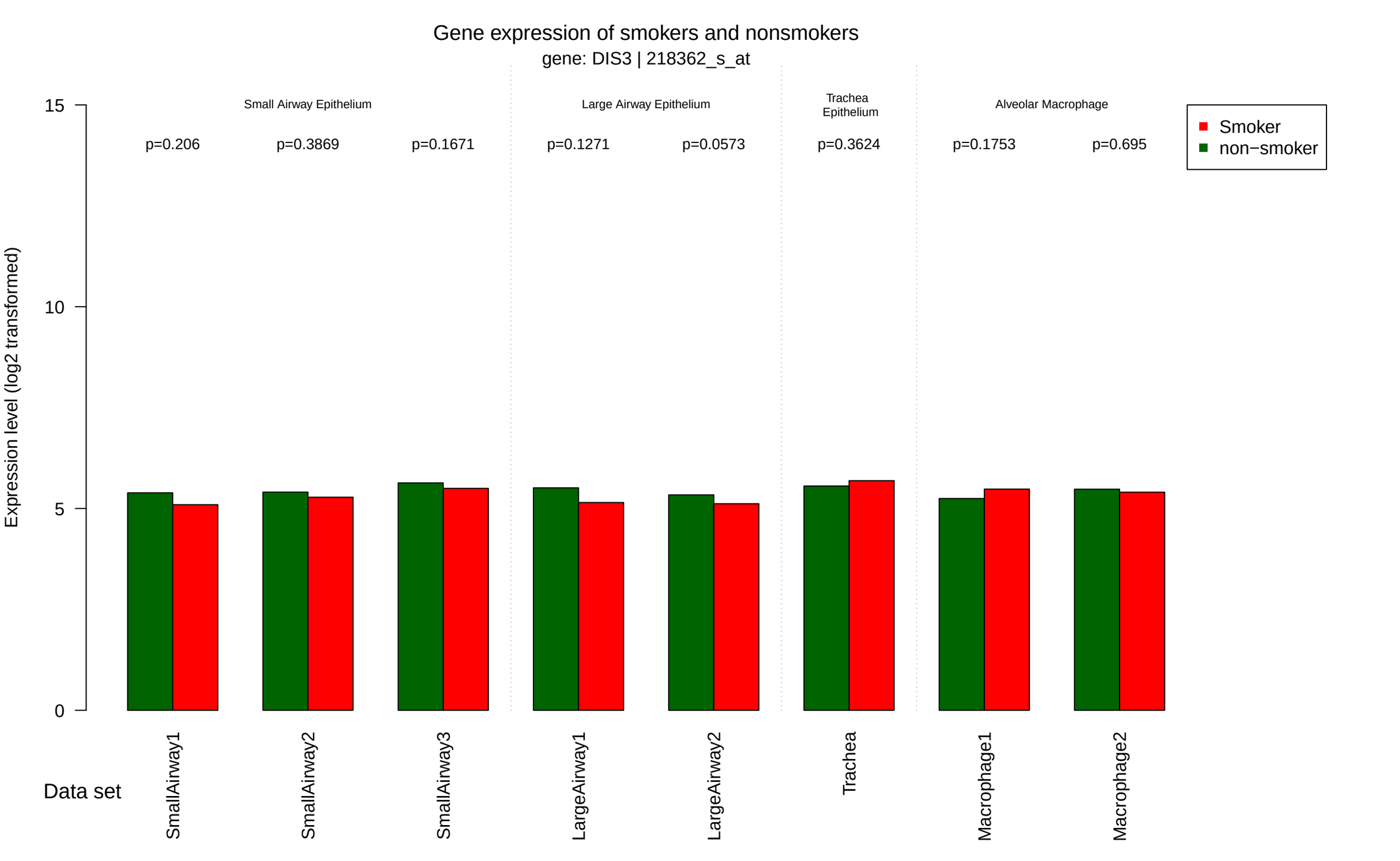
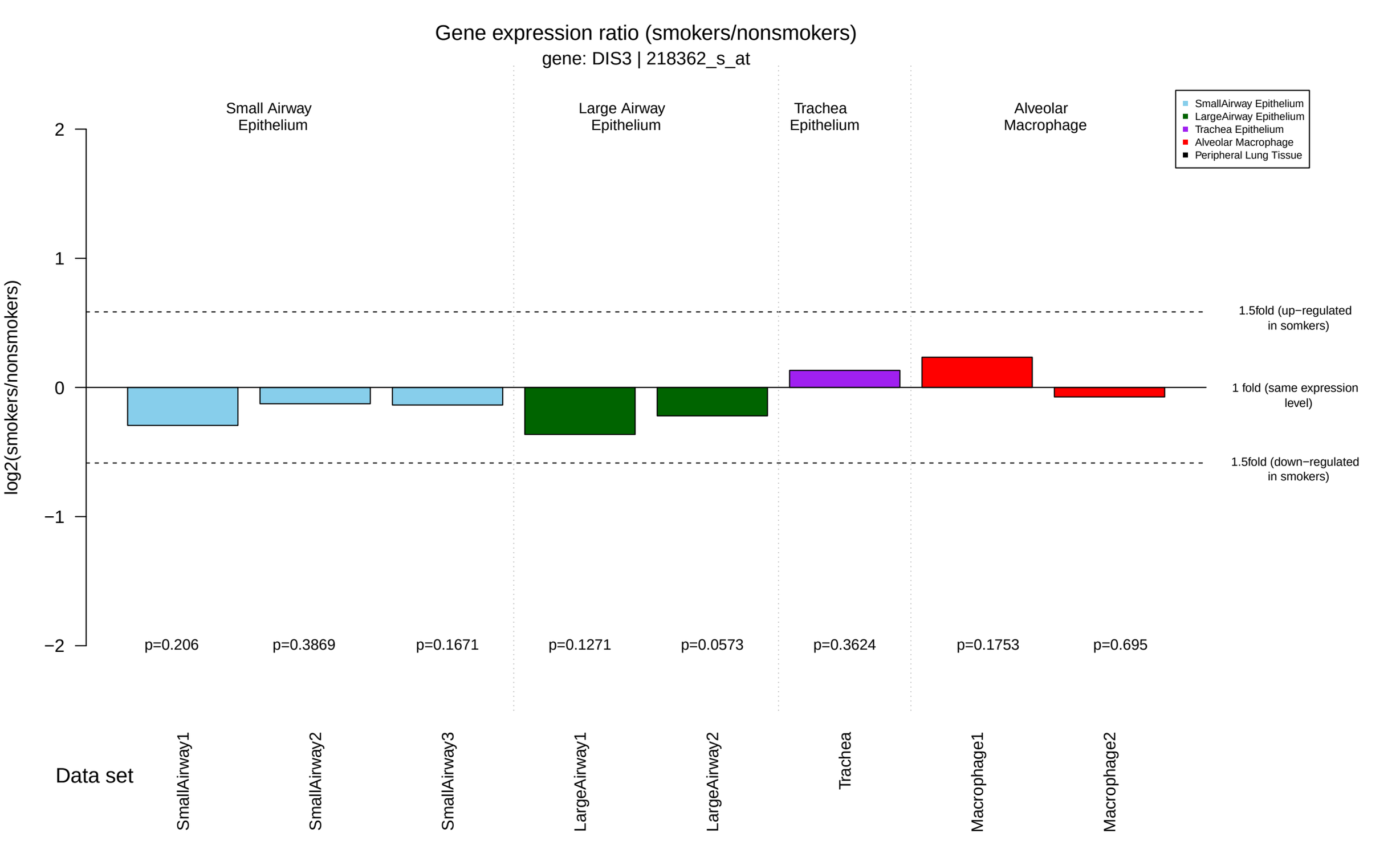
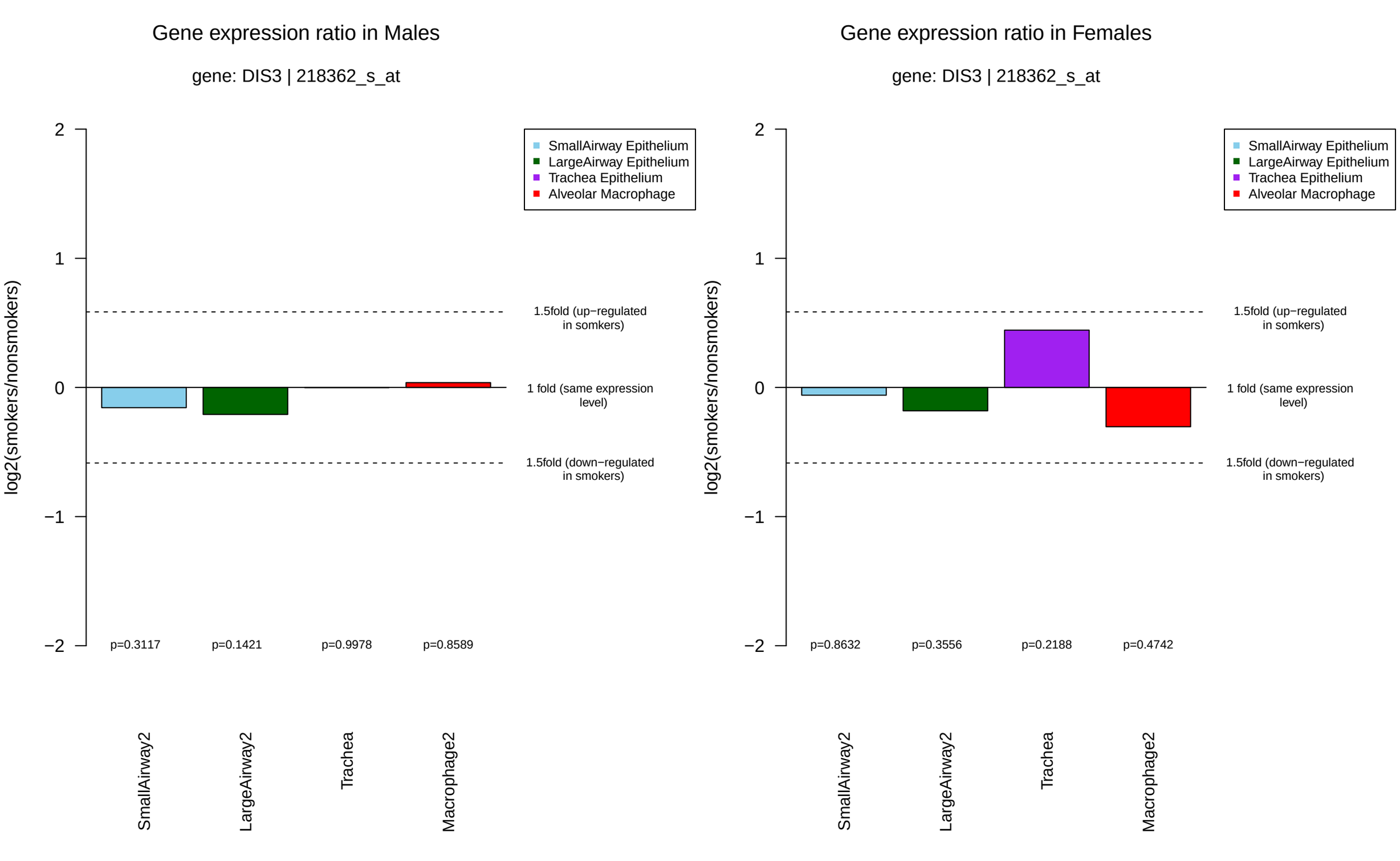
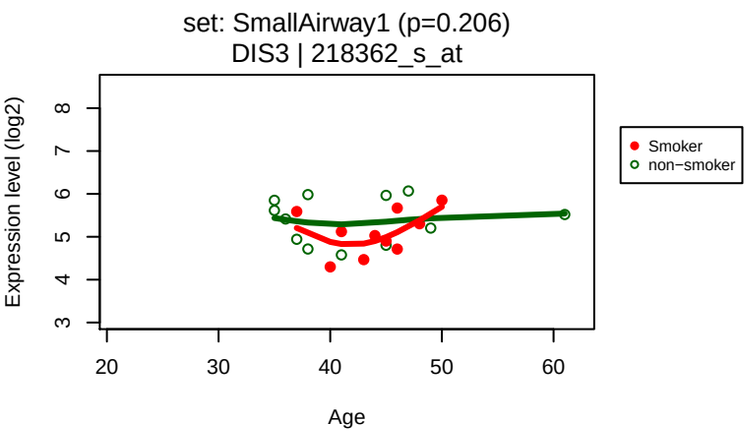
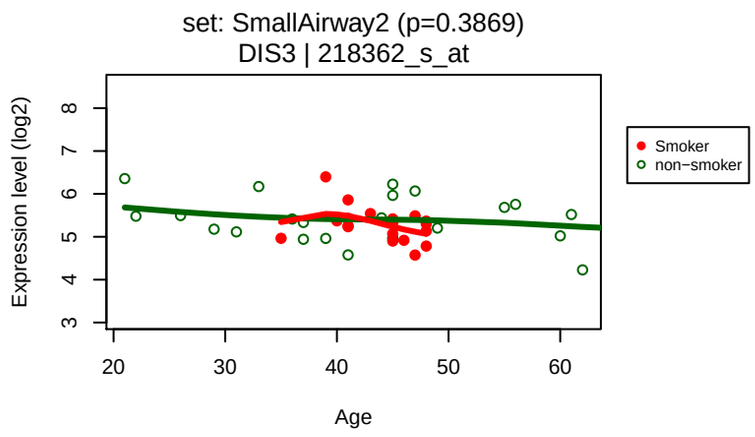
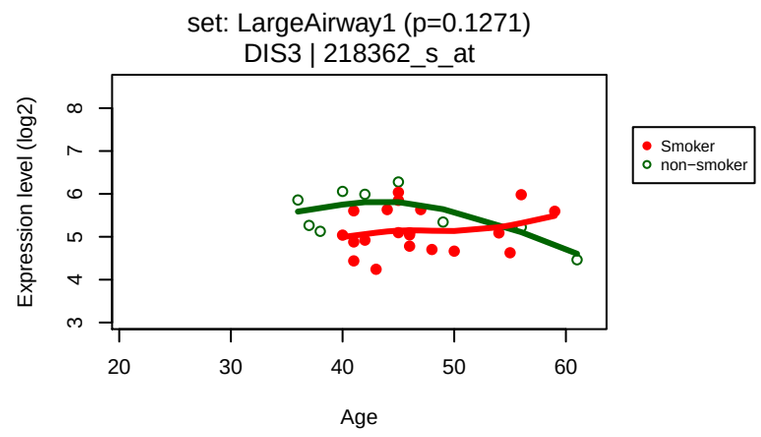

Gene expression of smokers and nonsmokers
gene: DIS3 | 218362_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.206
p=0.3869
p=0.1671
p=0.1271
p=0.0573
p=0.3624
p=0.1753
p=0.695
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

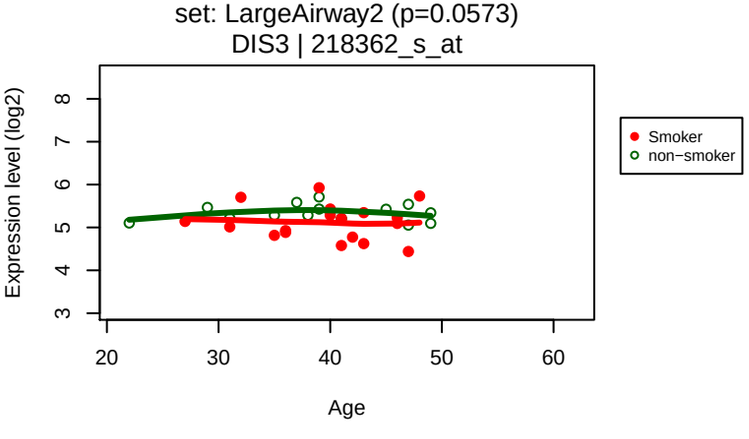

Gene expression ratio (smokers/nonsmokers)
gene: DIS3 | 218362_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.206
p=0.3869
p=0.1671
p=0.1271
p=0.0573
p=0.3624
p=0.1753
p=0.695
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

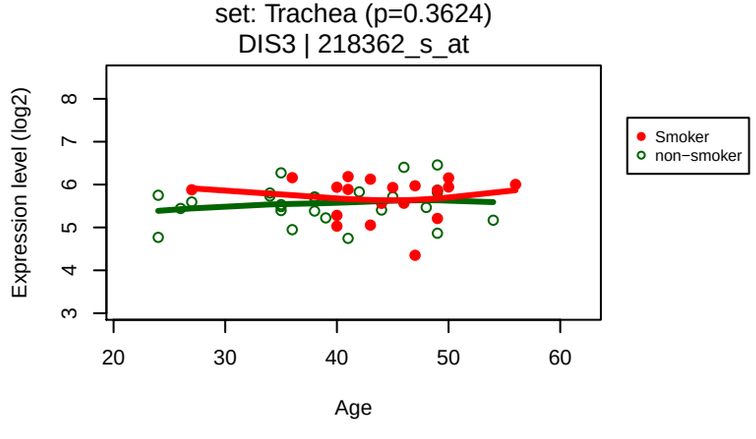

Gene expression ratio in Males
gene: DIS3 | 218362_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3117
p=0.1421
p=0.9978
p=0.8589
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DIS3 | 218362_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8632
p=0.3556
p=0.2188
p=0.4742
SmallAirway2
LargeAirway2
Trachea
Macrophage2

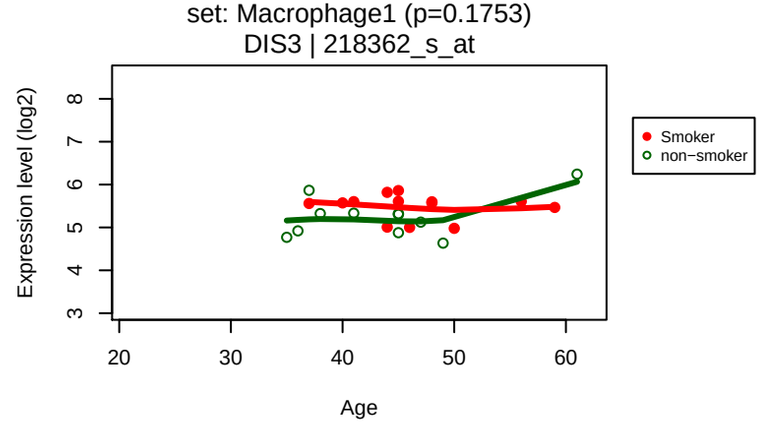

set: SmallAirway1 (p=0.206)
DIS3 | 218362_s_at
Expression level (log2)
Age
Smoker
non−smoker

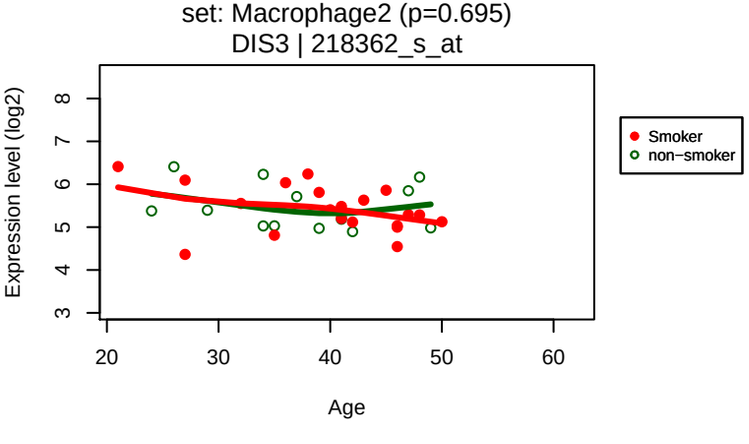

set: SmallAirway2 (p=0.3869)
DIS3 | 218362_s_at
Expression level (log2)
Age
Smoker
non−smoker

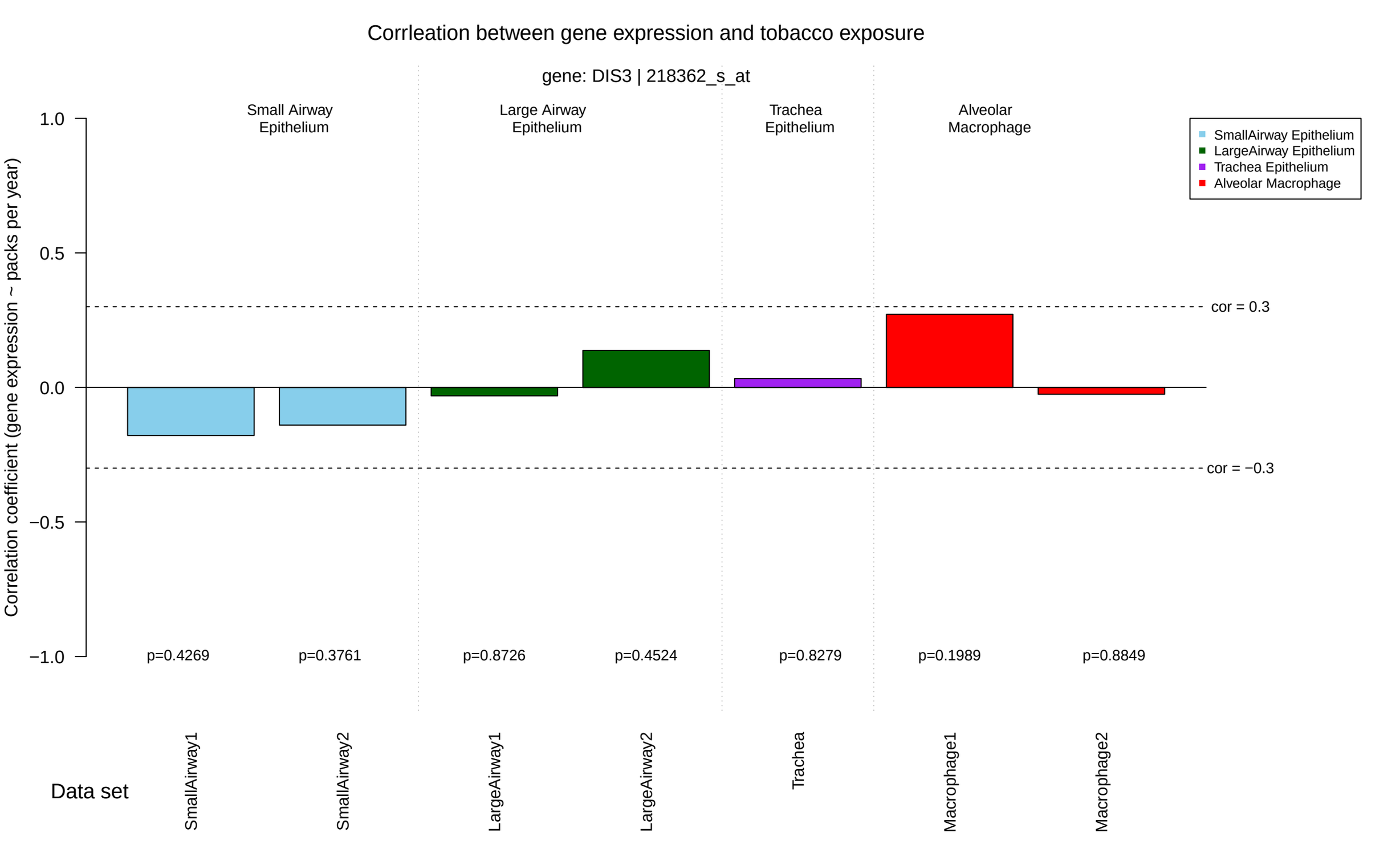

set: LargeAirway1 (p=0.1271)
DIS3 | 218362_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0573)
DIS3 | 218362_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3624)
DIS3 | 218362_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1753)
DIS3 | 218362_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.695)
DIS3 | 218362_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DIS3 | 218362_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4269
p=0.3761
p=0.8726
p=0.4524
p=0.8279
p=0.1989
p=0.8849
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2